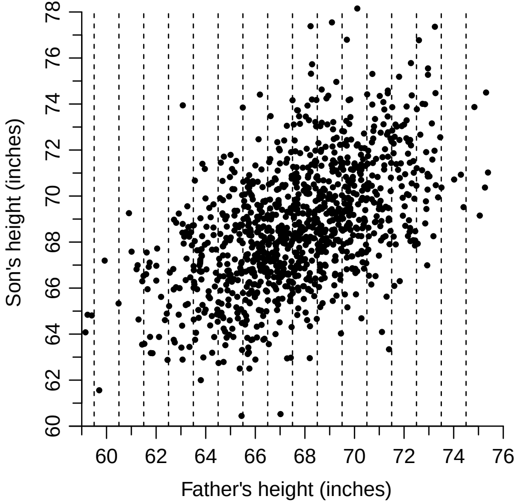
STAT 20000

Chapter 11 The R.M.S. Error for Regression

Chapter 12: The Regression Line

Example: Pearson's Father-and-Son Data

Fathers in the same vertical strip were equally tall, to the nearest inch.



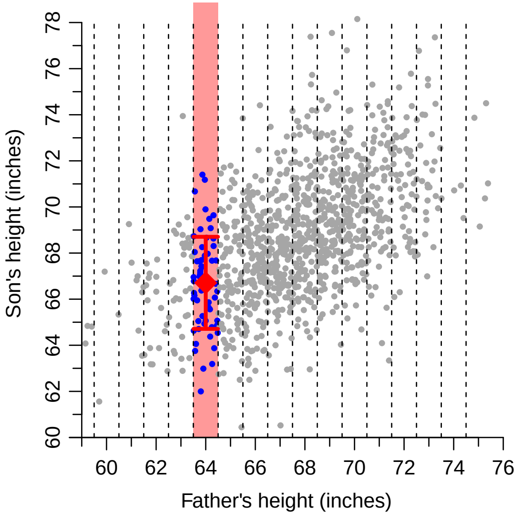
How do the

- mean of son's height (SH),
- SD of SH, and
- distribution of SH (histogram of SH)?

within each group change with father's height?

Example: Pearson's Father-and-Son Data

Fathers in the same vertical strip were equally tall, to the nearest inch.



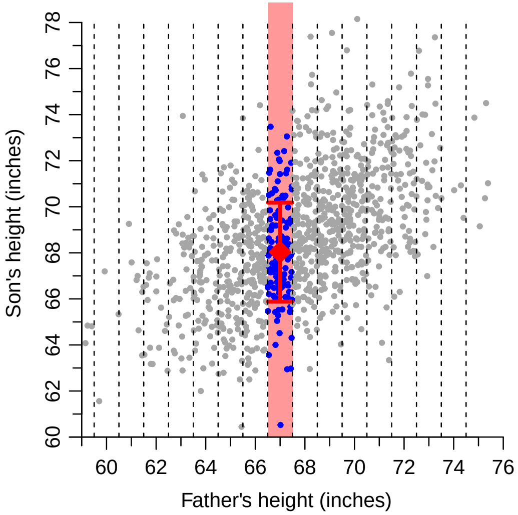
How do the

- mean of son's height (SH),
- SD of SH, and
- distribution of SH (histogram of SH)?

within each group change with father's height?

Example: Pearson's Father-and-Son Data

Fathers in the same vertical strip were equally tall, to the nearest inch.



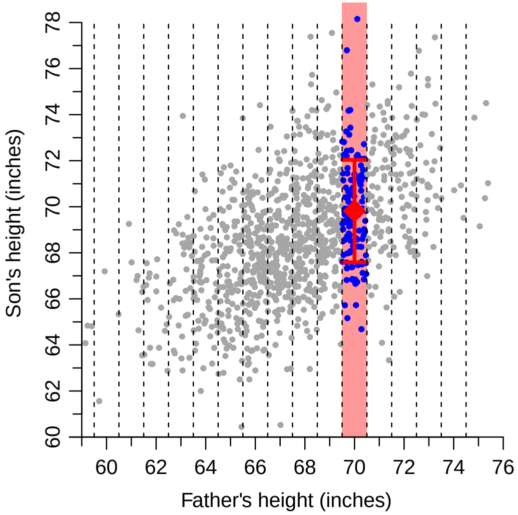
How do the

- mean of son's height (SH),
- SD of SH, and
- distribution of SH (histogram of SH)?

within each group change with father's height?

Example: Pearson's Father-and-Son Data

Fathers in the same vertical strip were equally tall, to the nearest inch.



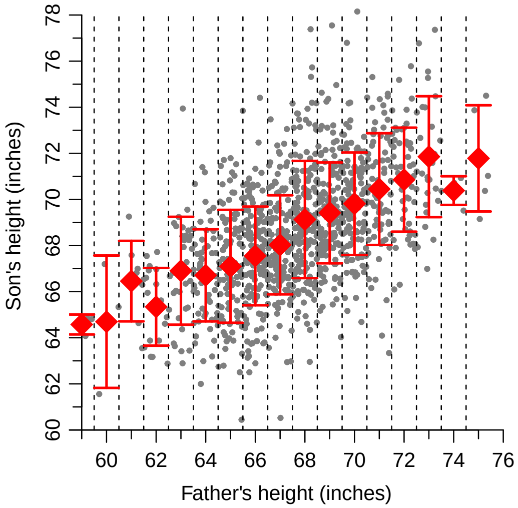
How do the

- mean of son's height (SH),
- SD of SH, and
- distribution of SH (histogram of SH)?

within each group change with father's height?

Example: Pearson's Father-and-Son Data

Fathers in the same vertical strip were equally tall, to the nearest inch.

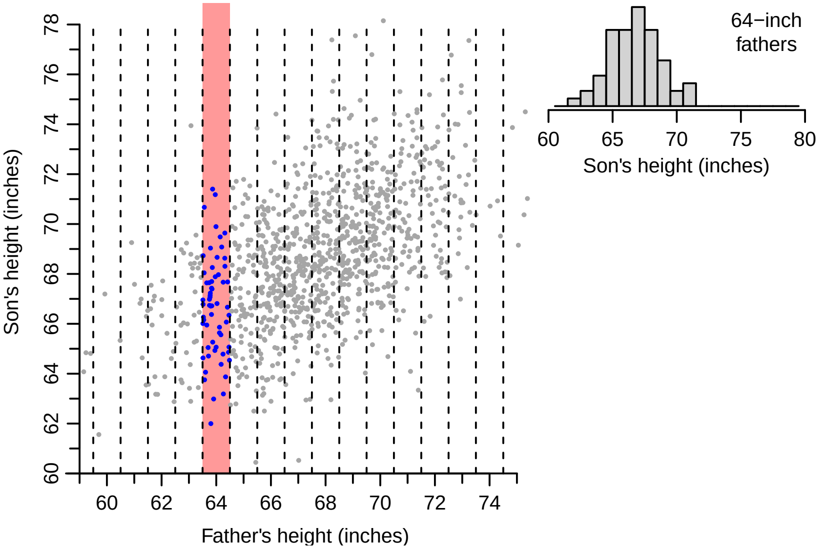


How do the

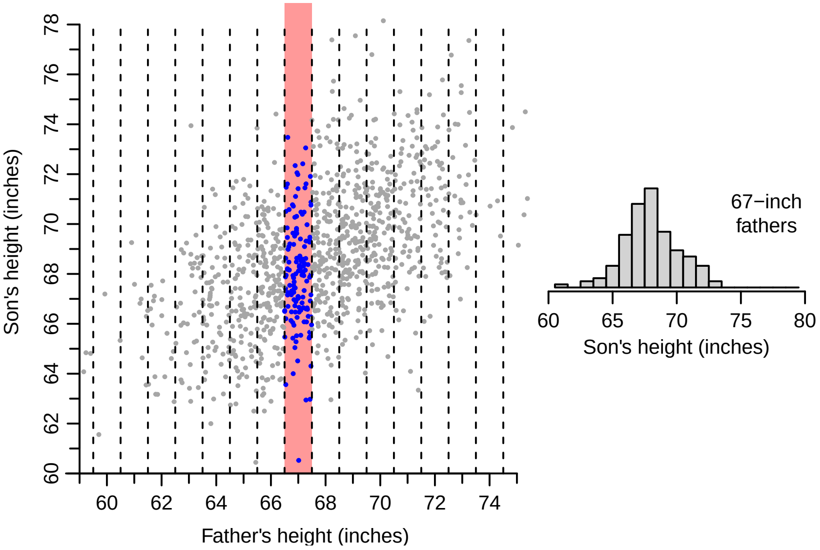
- mean of son's height (SH),
- SD of SH, and
- distribution of SH (histogram of SH)?

within each group change with father's height?

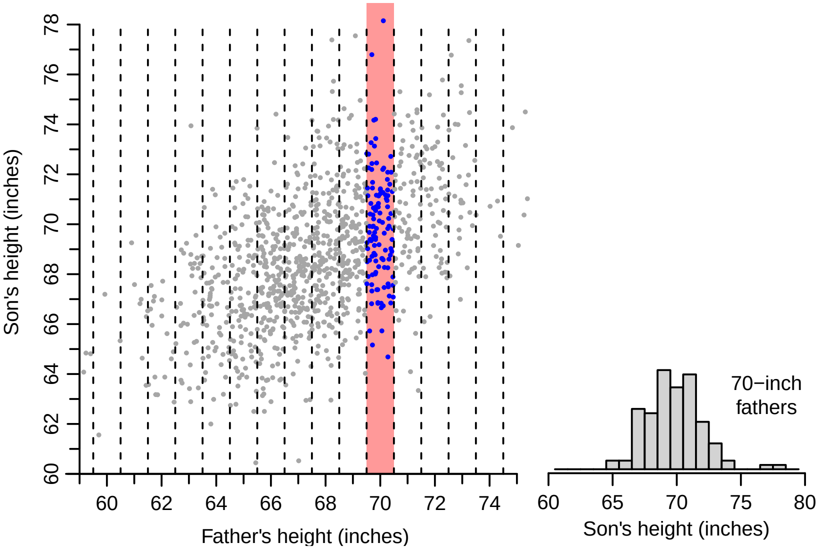
Example: Pearson's Father-and-Son Data



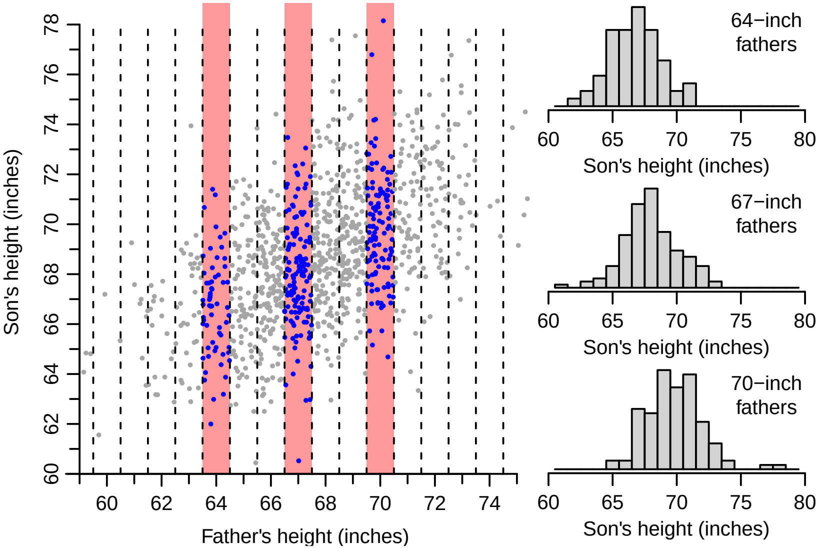
Example: Pearson's Father-and-Son Data



Example: Pearson's Father-and-Son Data



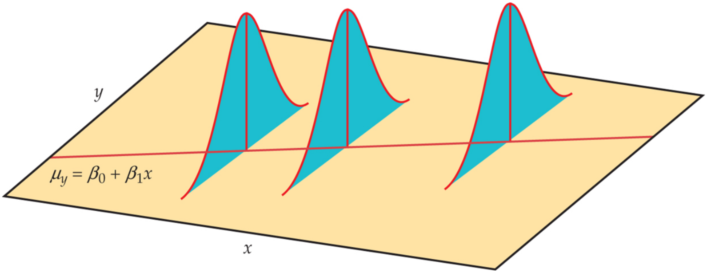
Example: Pearson's Father-and-Son Data



Assumptions of a Simple Linear Regression Model

Pearson's father-and-son data inspire the following assumptions for the simple linear regression (SLR) model:

1. The **conditional average** of Y given $X = x$ (i.e., the average of Y within the vertical strip at $X = x$) is linear in x
2. the **conditional SD** of Y given $X = x$ (i.e., the SD of Y within the vertical strip at x) is constant across different values of x
3. (Optional) the **conditional distribution** of Y given $X = x$ (i.e., the distribution of Y within the vertical strip at $X = x$) follows the normal curve



Outline

- Part I: Conditional mean (regression line)
  - Review of the equation of a straight line
  - Equation of the regression line
  - Interpretation of the regression equation
- Part II: Conditional SD (r.m.s. error)
  - Residual
  - Root-mean-square error (r.m.s. error)
  - Least-square method
  - The 68%- and 95%-rule for r.m.s. error
- Part III: Model diagnosis
  - After fitting a regression line, it's important to **check model assumptions** to verify that regression fit was OK.
  - Tool: Residual plot
  - Heteroscedasticity

Part I: Conditional Mean (Regression Line)

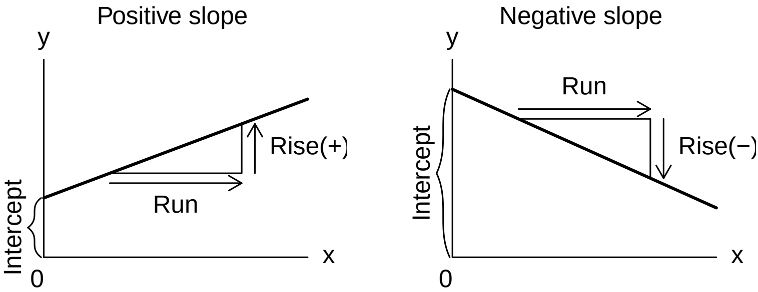
- Review of the equation of a straight line
- Equation of the regression line
- Interpretation of the regression equation

Equation of a Straight Line (Review)

The equation of a straight line is of the form

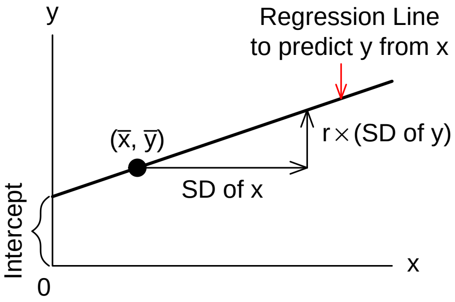
$$y = \text{intercept} + \text{slope} \times x$$

.



$$\text{Slope} = \frac{\text{Rise}}{\text{Run}}$$

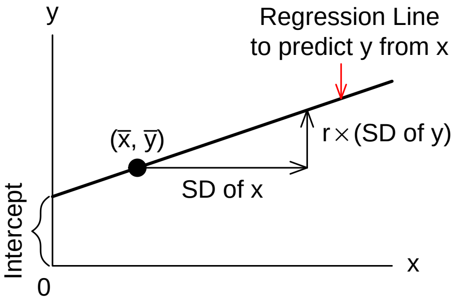
The Slope, Intercept, and Equation of a Regression Line



- equation: $y = \text{intercept} + \text{slope} \times x$.
- slope $=$ ______.
- intercept $=$ ______ $-$ __________ $\times$ ______.

With the equation, one can make regression predictions for several x's, by substituting each of them in turn into the equation.

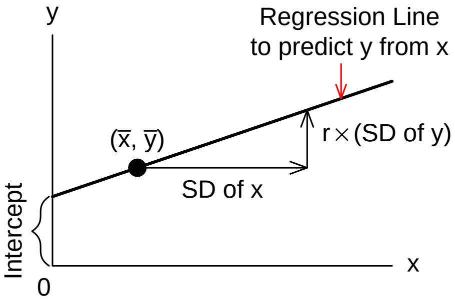
The Slope, Intercept, and Equation of a Regression Line



- equation: $y = \text{intercept} + \text{slope} \times x$.
- slope $= \dfrac{r\,\mathrm{SD}_y}{\mathrm{SD}_x}$.
- intercept $=$ _____ $-$ ________ $\times$ _____.

With the equation, one can make regression predictions for several x's, by substituting each of them in turn into the equation.

The Slope, Intercept, and Equation of a Regression Line



- equation: $y = \text{intercept} + \text{slope} \times x$.
- slope $= \dfrac{r\,\mathrm{SD}_y}{\mathrm{SD}_x}$.
- intercept $= \underline{\ \bar{y}\ } - \underline{\ \text{slope}\ } \times \underline{\ \bar{x}\ }$.

With the equation, one can make regression predictions for several x's, by substituting each of them in turn into the equation.

Example 1

In a sample of men age 18-24, the relationship between their heights and weights is summarized as follows

$$\begin{array}{rlll} \text{average height} \approx & 70", & \text{SD} \approx & 3" \\ \text{average weight} \approx & 162 \text{ lbs}, & \text{SD} \approx & 30 \text{ lbs}, \quad r \approx 0.5 \end{array}$$

The scatter diagram is approximately football-shaped. What is the regression equation for predicting height from weight?

- What is x? What is y?
- slope:
- intercept:
- equation:

Example 1

In a sample of men age 18-24, the relationship between their heights and weights is summarized as follows

$$\begin{array}{ll} \text{average height} \approx \;\; 70", & \text{SD} \approx \;\; 3" \\ \text{average weight} \approx 162 \text{ lbs}, & \text{SD} \approx 30 \text{ lbs} \;, \quad r \approx 0.5 \end{array}$$

The scatter diagram is approximately football-shaped. What is the regression equation for predicting height from weight?

- What is x? What is y? $x =$ weight, $y =$ height
- slope:
- intercept:
- equation:

Example 1

In a sample of men age 18-24, the relationship between their heights and weights is summarized as follows

$$\begin{array}{rlrl} \text{average height} \approx & 70", & \text{SD} \approx & 3" \\ \text{average weight} \approx & 162 \text{ lbs}, & \text{SD} \approx & 30 \text{ lbs}, \quad r \approx 0.5 \end{array}$$

The scatter diagram is approximately football-shaped. What is the regression equation for predicting height from weight?

- What is x? What is y? $x =$ weight, $y =$ height
- slope: $r\ SD_y/SD_x = 0.5 \times 3/30 = 0.05$
- intercept:
- equation:

Example 1

In a sample of men age 18-24, the relationship between their heights and weights is summarized as follows

$$\begin{array}{rll} \text{average height} \approx & 70", & SD \approx \ \ 3" \\ \text{average weight} \approx & 162 \text{ lbs}, & SD \approx 30 \text{ lbs}, \quad r \approx 0.5 \end{array}$$

The scatter diagram is approximately football-shaped. What is the regression equation for predicting height from weight?

- What is x? What is y? $x =$ weight, $y =$ height
- slope: $r\ SD_y/SD_x = 0.5 \times 3/30 = 0.05$
- intercept: $\bar{y} - (\text{slope}) \times \bar{x} = 70 - 0.05 \times 162 = 61.9$ lbs
- equation:

Example 1

In a sample of men age 18-24, the relationship between their heights and weights is summarized as follows

$$\begin{array}{rlr} \text{average height} \approx & 70", & \text{SD} \approx 3" \\ \text{average weight} \approx & 162 \text{ lbs}, & \text{SD} \approx 30 \text{ lbs}, \quad r \approx 0.5 \end{array}$$

The scatter diagram is approximately football-shaped. What is the regression equation for predicting height from weight?

- What is x? What is y? $x = \text{weight}$, $y = \text{height}$
- slope: $r\ SD_y/SD_x = 0.5 \times 3/30 = 0.05$
- intercept: $\bar{y} - (\text{slope}) \times \bar{x} = 70 - 0.05 \times 162 = 61.9$ lbs
- equation: $y = 61.9 + 0.05x$ or
  predicted height $= 61.9" + (0.05"\text{per lb}) \times (\text{weight})$.

Example 1 (Cont'd)

For men age 18-24, the regression line for predicting height from weight is

$$\text{predicted height} = 61.9" + (0.05"\text{per lb}) \times (\text{weight}).$$

On average, a man that weighs one more pound is 0.05" taller

- On average, a 160-pound man (age 18-24) will be 0.5" aller than a 150-pound man.
- John is 23 years old. If he puts on 10 pounds, will he become 0.5" taller? Explain.

- In this example, the intercept is meaningless since there is no man weighs 0 lbs.

Example 1 (Cont'd)

For men age 18-24, the regression line for predicting height from weight is

$$\text{predicted height} = 61.9" + (0.05"\text{per lb}) \times (\text{weight}).$$

On average, a man that weighs one more pound is 0.05" taller

- On average, a 160-pound man (age 18-24) will be 0.5" aller than a 150-pound man.
- John is 23 years old. If he puts on 10 pounds, will he become 0.5" taller? Explain.
  Clearly weight gain does not affect height. This study is observational. There are other (confounding) factors that effects both a man's height and weight.
- In this example, the intercept is meaningless since there is no man weighs 0 lbs.

Example 2

For a random sample of 1380 adult Americans in 1990, the regression equation for predicting annual earnings from height is

$$\text{predicted annual earning} = (\$1500 \text{ per inch}) \times (\text{height}) - \$80,000.$$

On average, an extra inch of height is worth \$1,500. How do you interpret this?

Example 2

For a random sample of 1380 adult Americans in 1990, the regression equation for predicting annual earnings from height is

$$\text{predicted annual earning} = (\$1500 \text{ per inch}) \times (\text{height}) - \$80,000.$$

On average, an extra inch of height is worth $1,500. How do you interpret this?

- This study is observational.
  There is at least one confounding variable – **gender**.
  Men are taller than women and tend to earn more than women.

Interpretation of the Slope & the Intercept

Lessons from the previous two examples:

With an observational study, the slope and intercept of the regression line are only descriptive statistics. They say how the average value of one variable is related to values of another variable, in the population being observed. The slope cannot be relied on to predict how y would respond if the investigator changed the value of x.

Statistical interpretation of the slope and the intercept:

- The slope is the **average change in y, associated with a unit increase in x.**
- If 0 is in the x-range of the data, then the intercept is the regression estimate for y when x is 0. In other cases, the intercept has no real meaning as a quantity.

Part II: Conditional SD (r.m.s. error)

- Residual
- Root-mean-square error (r.m.s. error)
- Least-square method
- The 68%- and 95%-rule for r.m.s. error

Residual (Prediction Error)

$$\begin{array}{cccccc} \text{Residual (Prediction Error)} & = & \text{Actual} & - & \text{Predicted} & \\ e_i & = & y_i & - & \widehat{y}_i & \text{(Notation)} \end{array}$$

Example:

Regression line:

- slope $= \dfrac{r\,\mathrm{SD}_y}{\mathrm{SD}_x} = \dfrac{0.4 \times 4}{2} = 0.8$
- intercept $= \bar{y} - (\text{slope}) \times \bar{x}$
  $= 7 - 0.8 \times 4 = 3.8$
- equation:

The predicted y_i:

$$\widehat{y}_i = 3.8 + 0.8x_i$$

Residual: $e_i = y_i - \widehat{y}_i$

x_i	y_i	Predicted $\hat{y}_i$	Residual $y_i - \hat{y}_i$
1	5	4.6	0.4
3	9	6.2	2.8
4	7	7.0	0.0
5	1	7.8	−6.8
7	13		

$\bar{x} = 4$, $\bar{y} = 7$,
$\mathrm{SD}_x = 2$, $\mathrm{SD}_y = 4$,
$r = 0.4$

Residual (Prediction Error)

$$\begin{array}{rcccc} \text{Residual (Prediction Error)} & = & \text{Actual} & - & \text{Predicted} \\ e_i & = & y_i & - & \widehat{y}_i \end{array} \qquad \text{(Notation)}$$

Example:

Regression line:

- slope $= \dfrac{r\,\mathrm{SD}_y}{\mathrm{SD}_x} = \dfrac{0.4 \times 4}{2} = 0.8$
- intercept $= \bar{y} - (\text{slope}) \times \bar{x}$ $= 7 - 0.8 \times 4 = 3.8$
- equation: $y = 3.8 + 0.8x$

The predicted y_i:

$$\widehat{y}_i = 3.8 + 0.8x_i$$

Residual: $e_i = y_i - \widehat{y}_i$

x_i	y_i	Predicted $\hat{y}_i$	Residual $y_i - \hat{y}_i$
1	5	4.6	0.4
3	9	6.2	2.8
4	7	7.0	0.0
5	1	7.8	−6.8
7	13		

$\bar{x} = 4$, $\bar{y} = 7$,
$\mathrm{SD}_x = 2$, $\mathrm{SD}_y = 4$,
$r = 0.4$

Residual (Prediction Error)

$$\begin{array}{lclcl} \text{Residual (Prediction Error)} & = & \text{Actual} & - & \text{Predicted} \\ e_i & = & y_i & - & \widehat{y}_i \end{array} \qquad \text{(Notation)}$$

Example:

Regression line:

- slope $= \dfrac{r\,\mathrm{SD}_y}{\mathrm{SD}_x} = \dfrac{0.4 \times 4}{2} = 0.8$
- intercept $= \bar{y} - (\text{slope}) \times \bar{x}$
  $= 7 - 0.8 \times 4 = 3.8$
- equation: $y = 3.8 + 0.8x$

The predicted y_i:

$$\widehat{y}_i = 3.8 + 0.8x_i$$

Residual: $e_i = y_i - \widehat{y}_i$

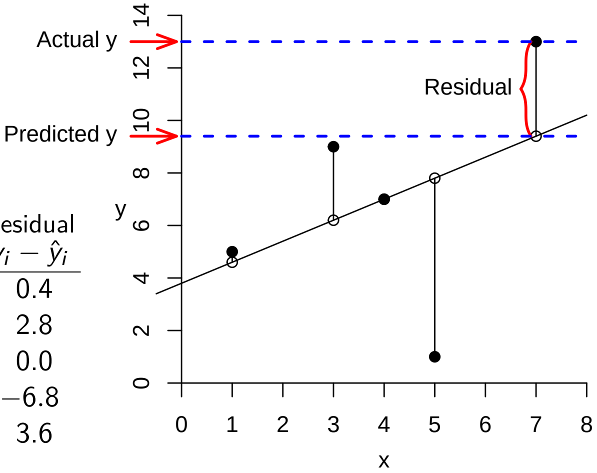
x_i	y_i	Predicted $\hat{y}_i$	Residual $y_i - \hat{y}_i$
1	5	4.6	0.4
3	9	6.2	2.8
4	7	7.0	0.0
5	1	7.8	−6.8
7	13	9.4	3.6

$\bar{x} = 4$, $\bar{y} = 7$,
$\mathrm{SD}_x = 2$, $\mathrm{SD}_y = 4$,
$r = 0.4$

Residual (Prediction Error)

$$\text{Residual (Prediction Error)} = \text{Actual } y - \text{Predicted } y$$

x_i	Actual y_i	Predicted $\hat{y}_i$	Residual $y_i - \hat{y}_i$
1	5	4.6	0.4
3	9	6.2	2.8
4	7	7.0	0.0
5	1	7.8	-6.8
7	13	9.4	3.6



The R.M.S. Error of Prediction

r.m.s. error = r.m.s. of the residuals

$$= \sqrt{\frac{(\text{Residual }1)^2 + (\text{Residual }2)^2 + \ldots + (\text{Residual n})^2}{n}}$$

- measures the overall size of prediction errors.
- estimate the **conditional SD of** y **given** x

In the former example,

x	y	Predicted y	Residual	$(\text{Residual})^2$
1	5	4.6	0.4	0.16
3	9	6.2	2.8	7.84
4	7	7.0	0.0	0.00
5	1	7.8	−6.8	46.24
7	13	9.4	3.6	12.96

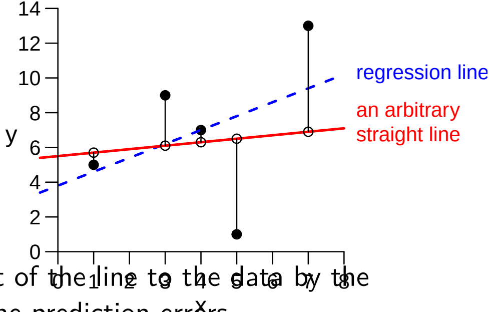
$$\text{r.m.s. error} = \sqrt{\frac{0.16 + 7.84 + 0 + 46.24 + 12.96}{5}} = \sqrt{13.44} = 3.67$$

Technical Notes: The Method of Least Squares

Using an arbitrary straight line $y = a + bx$ to predict y from x

predicted $y_i = a + bx_i$,

the prediction errors are
$$y_i - a - bx_i.$$

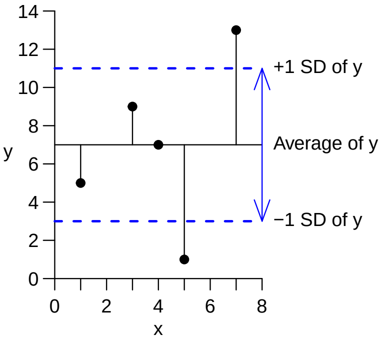


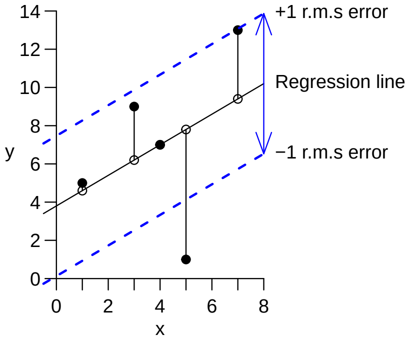
- Measure the lack of fit of the line to the data by the root-mean-square of the prediction errors
$$\text{r.m.s. error} = \sqrt{\sum_{i=1}^{n}(y_i - a - bx_i)^2/n}$$
- The regression line is the line with the **smallest** r.m.s. error among all straight lines, and hence is the **least-square line**.

SD versus R.M.S. Error

SD is the r.m.s. of the deviations from the average $(y_i - \bar{y})$;

r.m.s. error is the r.m.s. of the residuals $(y_i - \hat{y}_i)$



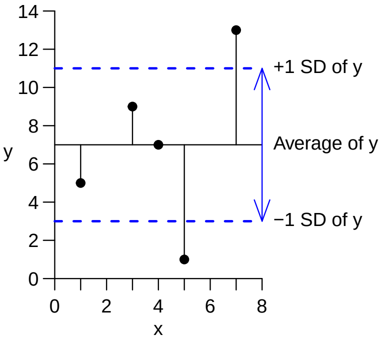


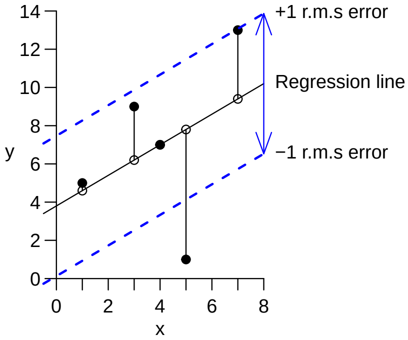
Which one do you think is smaller, the SD of y or the r.m.s. error of the prediction of y based on regression on x?

SD versus R.M.S. Error

SD is the r.m.s. of the deviations from the average $(y_i - \bar{y})$;

r.m.s. error is the r.m.s. of the residuals $(y_i - \hat{y}_i)$





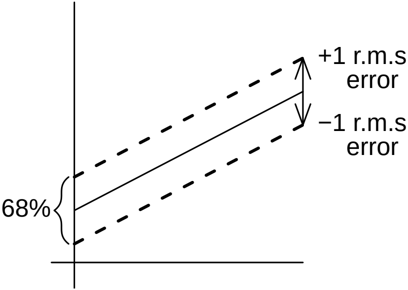
Which one do you think is smaller, the SD of y or the r.m.s. error of the prediction of y based on regression on x? r.m.s. error.

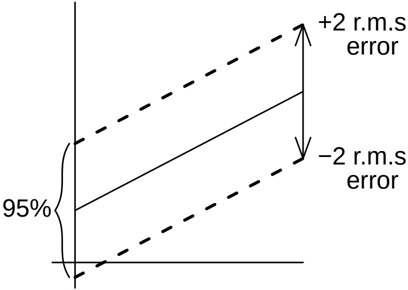
The 68%- and 95%-Rule for R.M.S. Error

Do you remember the 68%- and 95%-rules for SD?

If a regression line for predicting test scores has a r.m.s. error of 8 points, then

- About 68% of the time, the predictions will be right to within _____ points.
- About 95% of the time, the predictions will be right to within ______ points.



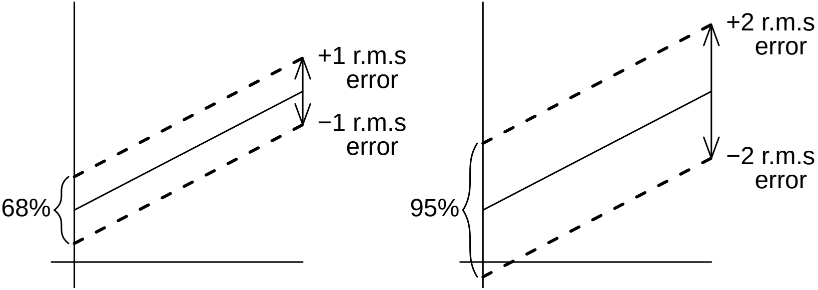


The 68%- and 95%-Rule for R.M.S. Error

Do you remember the 68%- and 95%-rules for SD?

If a regression line for predicting test scores has a r.m.s. error of 8 points, then

- About 68% of the time, the predictions will be right to within __8__ points.
- About 95% of the time, the predictions will be right to within ______ points.

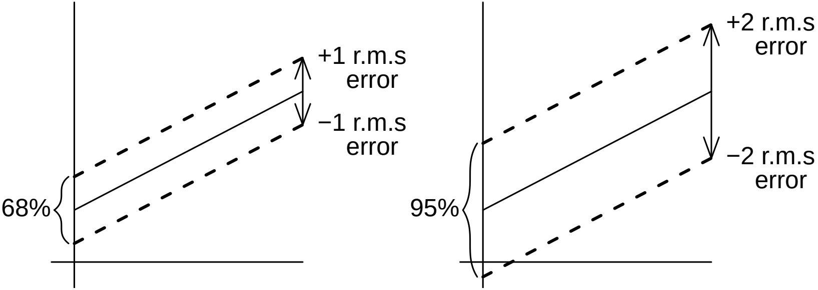


The 68%- and 95%-Rule for R.M.S. Error

Do you remember the 68%- and 95%-rules for SD?

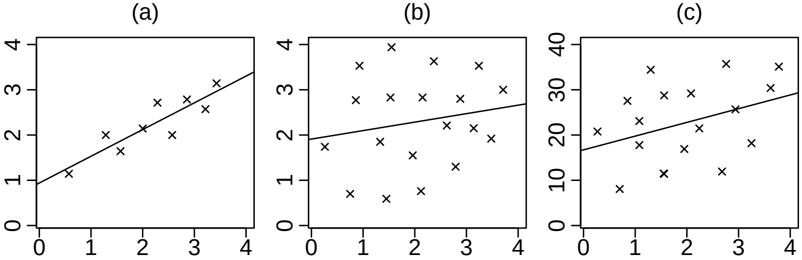
If a regression line for predicting test scores has a r.m.s. error of 8 points, then

- About 68% of the time, the predictions will be right to within __8__ points.
- About 95% of the time, the predictions will be right to within __16__ points.



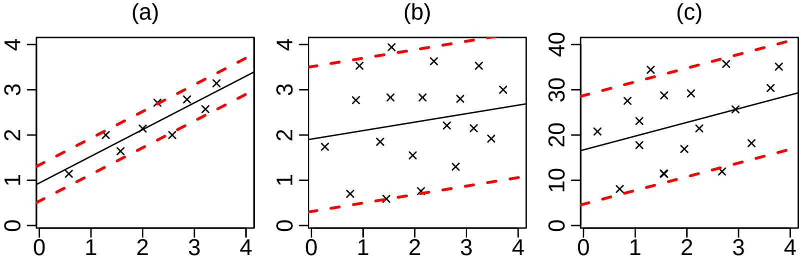
Exercise

Below are 3 scatter diagrams. The regression line has been drawn across each one. In each case, guess whether the r.m.s. error is about 0.2, 1, or 5.



Exercise

Below are 3 scatter diagrams. The regression line has been drawn across each one. In each case, guess whether the r.m.s. error is about 0.2, 1, or 5.



(a)	(b)	(c)
0.2	1	5

Shortcut Formula for the R.M.S. Error

The r.m.s. error for the regression line (of y on x) is related to the SD of y and the correlation coefficient r from the formula,

$$\text{r.m.s. error} = \sqrt{1 - r^2} \times (\text{the SD of } y)$$

Point	x	y
A	1	5
B	3	7
C	4	9
D	5	1
E	7	13
Ave	4	7
SD	2	4

$$\begin{aligned} r &= 0.4 \\ \text{SD of } y &= 4 \\ \text{r.m.s. error} &= \sqrt{1 - r^2} \times (\text{SD of } y) \\ &= \sqrt{1 - (0.4)^2} \times 4 \\ &= 3.67 \end{aligned}$$

The shortcut formula works for all scatter diagrams, but is interpretable/useful only for football-shaped.

More about the R.M.S. Error

- If you didn't know x, you'd predict y by the overall average of the y's. In this case your prediction would be off by amounts similar in size to the __________
- Predictions based on the regression line make use of the information about y conveyed by x, and are more accurate than predictions made without using x by a factor of $\sqrt{1 - r^2}$.

Q: What are the units for the r.m.s. error?

A: Same as y;
that's what reminds you to multiply by the SD of y.

Q: Suppose you want to predict some "response" variable y. There are several "explanatory" variables x you might use, but you can only use one of them. Which one would you pick, and why?

More about the R.M.S. Error

- If you didn't know x, you'd predict y by the overall average of the y's. In this case your prediction would be off by amounts similar in size to the SD of y
- Predictions based on the regression line make use of the information about y conveyed by x, and are more accurate than predictions made without using x by a factor of $\sqrt{1-r^2}$.

Q: What are the units for the r.m.s. error?

A: Same as y;
that's what reminds you to multiply by the SD of y.

Q: Suppose you want to predict some "response" variable y. There are several "explanatory" variables x you might use, but you can only use one of them. Which one would you pick, and why?

More about the R.M.S. Error

- If you didn't know x, you'd predict y by the overall average of the y's. In this case your prediction would be off by amounts similar in size to the SD of y
- Predictions based on the regression line make use of the information about y conveyed by x, and are more accurate than predictions made without using x by a factor of $\sqrt{1 - r^2}$.

Q: What are the units for the r.m.s. error?

A: Same as y;
that's what reminds you to multiply by the SD of y.

Q: Suppose you want to predict some "response" variable y. There are several "explanatory" variables x you might use, but you can only use one of them. Which one would you pick, and why?

A: The one with the largest $|r|$.

Example: Twins

In one study of identical male twins, the average height was found to be about 68", with an SD of about 3". The correlation between the heights of the twins was about 0.95, and the scatter diagram was football-shaped.

(a) What method would you use to guess the height of one research subject, without any further information?
method =

(b) Find the r.m.s. error for the method in (a).
r.m.s. error =

(c) If one of the twin is known to be 6' 2"tall, what method would you use to guess the height of the other twin?
method =
height of the other twin =

(d) Find the r.m.s. error for the method in (c).
r.m.s. error =

Example: Twins

In one study of identical male twins, the average height was found to be about 68", with an SD of about 3". The correlation between the heights of the twins was about 0.95, and the scatter diagram was football-shaped.

(a) What method would you use to guess the height of one research subject, without any further information?
method = average = 68"

(b) Find the r.m.s. error for the method in (a).
r.m.s. error =

(c) If one of the twin is known to be 6' 2"tall, what method would you use to guess the height of the other twin?
method =
height of the other twin =

(d) Find the r.m.s. error for the method in (c).
r.m.s. error =

Example: Twins

In one study of identical male twins, the average height was found to be about 68", with an SD of about 3". The correlation between the heights of the twins was about 0.95, and the scatter diagram was football-shaped.

(a) What method would you use to guess the height of one research subject, without any further information?
method = average = 68"

(b) Find the r.m.s. error for the method in (a).
r.m.s. error = SD of y = 3"

(c) If one of the twin is known to be 6' 2"tall, what method would you use to guess the height of the other twin?
method =
height of the other twin =

(d) Find the r.m.s. error for the method in (c).
r.m.s. error =

Example: Twins

In one study of identical male twins, the average height was found to be about 68", with an SD of about 3". The correlation between the heights of the twins was about 0.95, and the scatter diagram was football-shaped.

(a) What method would you use to guess the height of one research subject, without any further information?
method = average = 68"

(b) Find the r.m.s. error for the method in (a).
r.m.s. error = SD of y = 3"

(c) If one of the twin is known to be 6' 2"tall, what method would you use to guess the height of the other twin?
method = regression
height of the other twin =

(d) Find the r.m.s. error for the method in (c).
r.m.s. error =

Example: Twins

In one study of identical male twins, the average height was found to be about 68", with an SD of about 3". The correlation between the heights of the twins was about 0.95, and the scatter diagram was football-shaped.

(a) What method would you use to guess the height of one research subject, without any further information?
method = average = 68"

(b) Find the r.m.s. error for the method in (a).
r.m.s. error = SD of y = 3"

(c) If one of the twin is known to be 6' 2"tall, what method would you use to guess the height of the other twin?
method = regression
height of the other twin = (Ave of y)+ $rSD_y \times SU_x$
$= 68" + 0.95 \times 3" \times 2 = 68" + 5.4" = 73.4"$

(d) Find the r.m.s. error for the method in (c).
r.m.s. error =

Example: Twins

In one study of identical male twins, the average height was found to be about 68", with an SD of about 3". The correlation between the heights of the twins was about 0.95, and the scatter diagram was football-shaped.

(a) What method would you use to guess the height of one research subject, without any further information?
method = average = 68"

(b) Find the r.m.s. error for the method in (a).
r.m.s. error = SD of y = 3"

(c) If one of the twin is known to be 6' 2"tall, what method would you use to guess the height of the other twin?
method = regression
height of the other twin = (Ave of y)+ $rSD_y \times SU_x$
$= 68" + 0.95 \times 3" \times 2 = 68" + 5.4" = 73.4"$

(d) Find the r.m.s. error for the method in (c).
r.m.s. error = $\sqrt{1 - r^2}\,\text{SD}_y = \sqrt{0.0975} \times 3" = 0.94"$

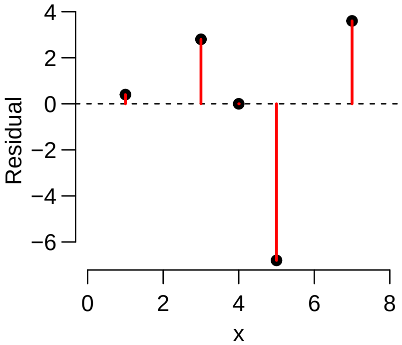
Part III: Model Diagnosis

- After fitting a regression line it is important to **check model assumptions** to verify that regression fit was OK.
- Residual Plots
- Homoscedasticity/Heteroscedasticity

Residual Plots

A residual plot is a plot of the residuals versus x, or perhaps versus some other relevant variable.

x	y	Residual
1	5	0.4
3	9	2.8
4	7	0.0
5	1	−6.8
7	13	3.6



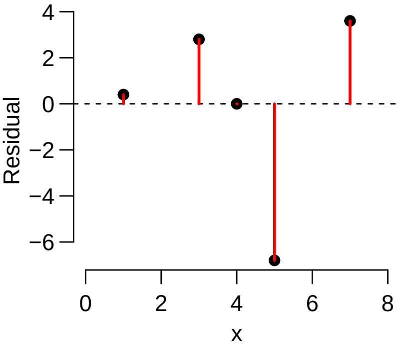
In general, the residuals average out to _____ , and the regression line for the residual plot is _______________.

- All the linear trend in the data is accounted for by the regression line for the data.

Residual Plots

A residual plot is a plot of the residuals versus x, or perhaps versus some other relevant variable.

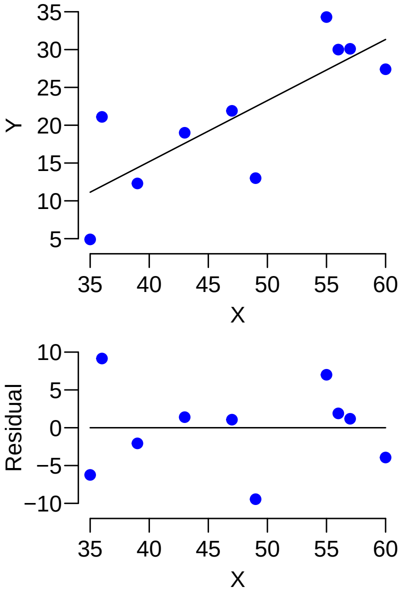
x	y	Residual
1	5	0.4
3	9	2.8
4	7	0.0
5	1	−6.8
7	13	3.6



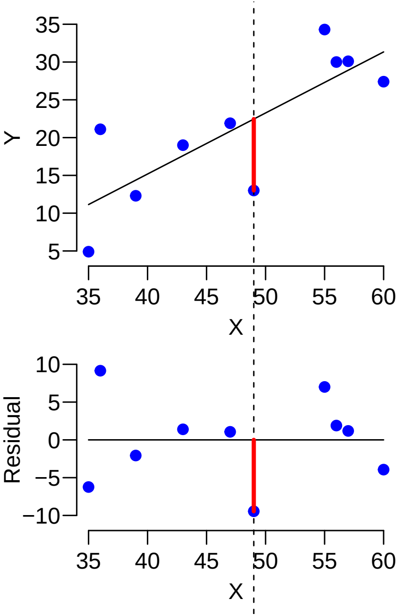
In general, the residuals average out to __0__, and the regression line for the residual plot is __the zero line__.

- All the linear trend in the data is accounted for by the regression line for the data.

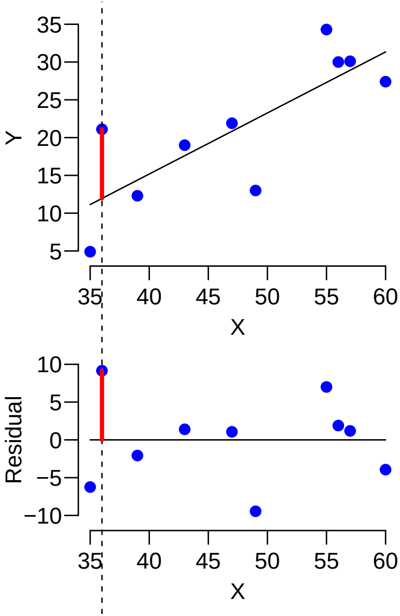
Relation Between Residual Plots and ScatterPlots



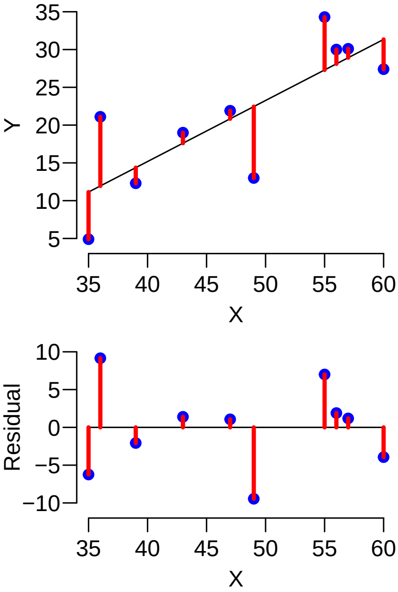
Relation Between Residual Plots and ScatterPlots



Relation Between Residual Plots and ScatterPlots



Relation Between Residual Plots and ScatterPlots



Using Residual Plots for Model Diagnosis

In an ideal residual plot, the data should spread evenly around the zero line. If there are any

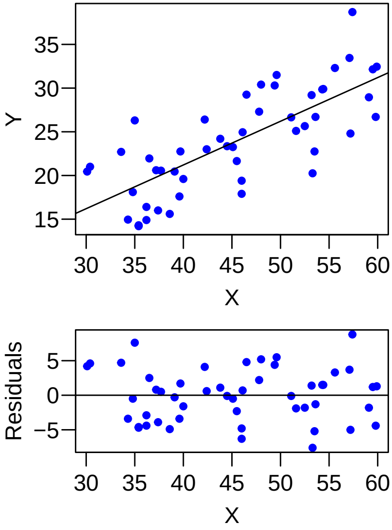
- outliers,
- unusual patterns,
- other signs that it would be a mistake to make predictions using a straight line,

then the simple linear regression model is not appropriate to describe the data. One should adjust the model or look for other models instead.

If Model Assumptions Are Satisfied . . .

If linearity and constant variability conditions are satisfied, points should **scatter evenly above and below**

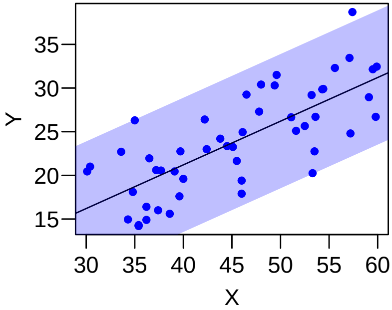
- the **regression line** in the scatter plot, and
- the **zero line** in the residual plot.

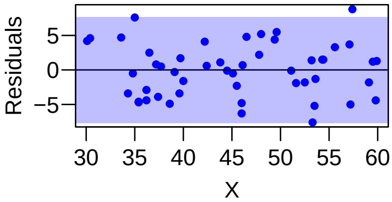


If Model Assumptions Are Satisfied . . .

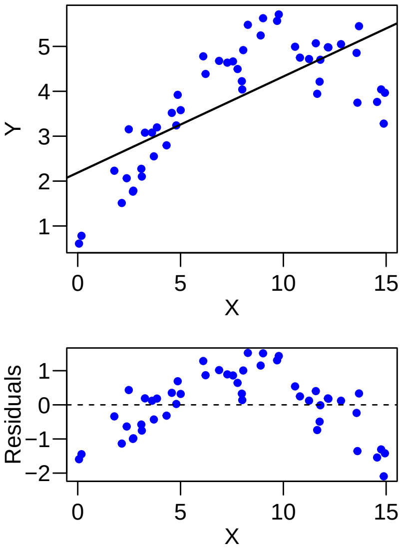
If linearity and constant variability conditions are satisfied, points should **scatter evenly above and below**

- the **regression line** in the scatter plot, and
- the **zero line** in the residual plot.

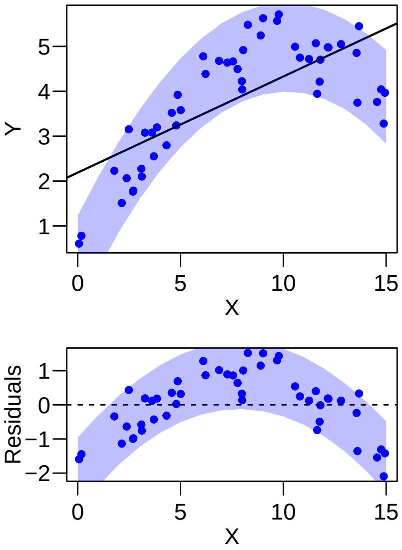




Violations of Linearity



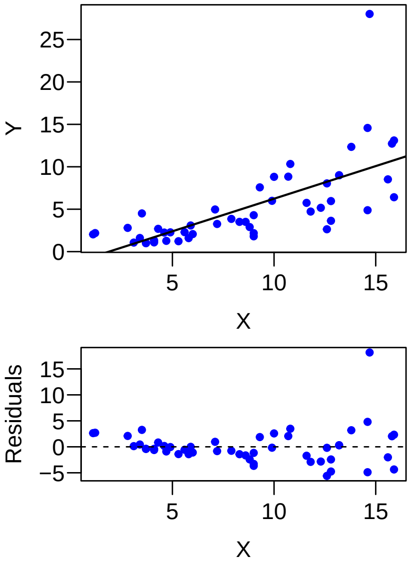
Violations of Linearity



Violations of Constant Variability

A scatter diagram or residual plot is called **homoscedastic** if all the vertical strips in the diagram show similar amounts of spread. Otherwise, it is **heteroscedastic.** (*Homo* means "same", *hetero* means "different" *scedastic* means "scatter".)

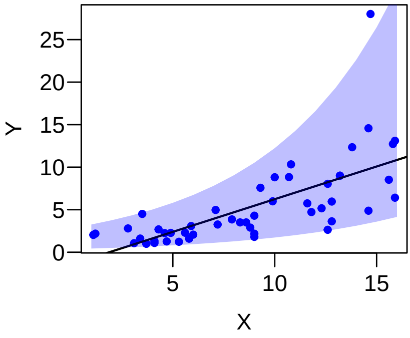
If a scatter diagram shows heteroscedasticity, the prediction errors changes with X of the scatter diagram. In that case, the **r.m.s. error** would NOT be as an universal estimate for the size of the prediction error.

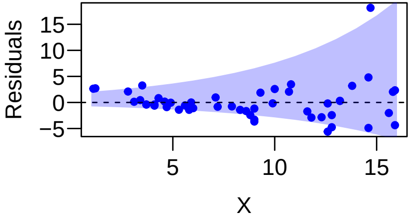


Violations of Constant Variability

A scatter diagram or residual plot is called **homoscedastic** if all the vertical strips in the diagram show similar amounts of spread. Otherwise, it is **heteroscedastic.** (*Homo* means "same", *hetero* means "different" *scedastic* means "scatter".)

If a scatter diagram shows heteroscedasticity, the prediction errors changes with X of the scatter diagram. In that case, the **r.m.s. error** would NOT be as an universal estimate for the size of the prediction error.



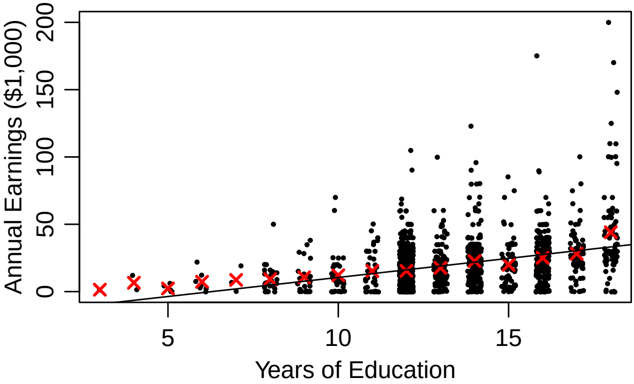


An Example of Heteroscedasticity

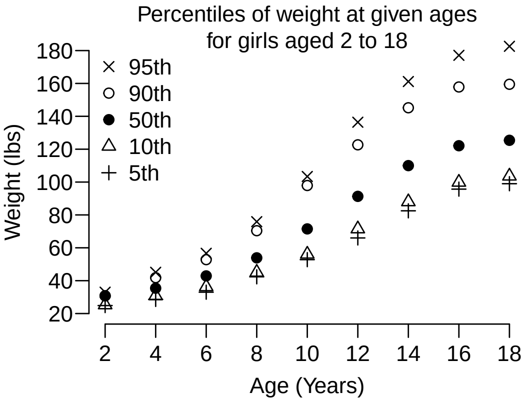
In a random sample of 1380 adult Americans in 1990, the regression equation for predicting annual earnings from education is found to be

$$\text{predicted annual earning} = (\$2,832 \text{ per year}) \times (\text{education}) - \$17,807.$$

This is, however, problematic, since ...



Another Example of Heteroscedasticity

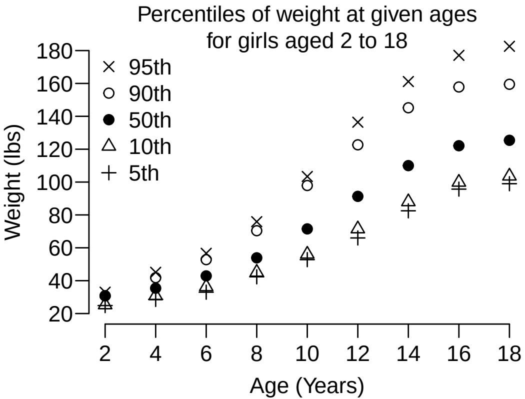


How does weight vary with respect to age?

- In regards to center?
- In regards to spread?

Is the distribution of weight at age 18 normal, or non-normal?

Another Example of Heteroscedasticity

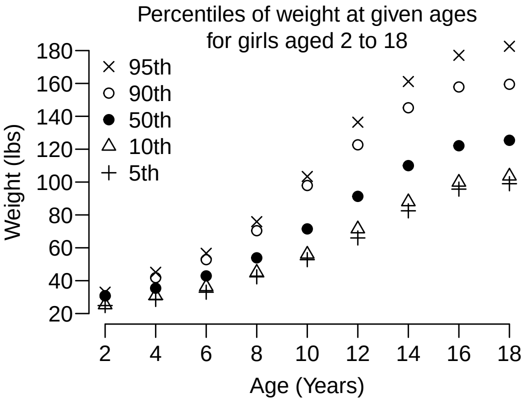


How does weight vary with respect to age?

- In regards to center? Increased with age **nonlinearly**.
- In regards to spread?

Is the distribution of weight at age 18 normal, or non-normal?

Another Example of Heteroscedasticity

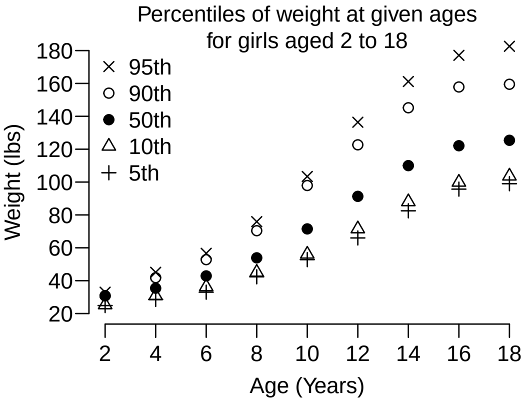


How does weight vary with respect to age?

- In regards to center? Increased with age **nonlinearly**.
- In regards to spread? Increased with age

Is the distribution of weight at age 18 normal, or non-normal?

Another Example of Heteroscedasticity



How does weight vary with respect to age?

- In regards to center? Increased with age **nonlinearly**.
- In regards to spread? Increased with age

Is the distribution of weight at age 18 normal, or non-normal?
Looks right-skewed. Median is left of center.

Summary

- With a regression line the difference between the actual value and the predicted value is called a *residual*.
- The RMS error measures the "overall size" of the residuals.
- The RMS error for the regression line of y on x can be figured as
$$\sqrt{1 - r^2} \times \text{SD of } y$$
- The residual plot shows the residuals! If a pattern appears then the linear regression may not have captured all the structure in the data set.
- Data sets with similar amounts of spread in all vertical strips are called *homoscedastic*. Those that show different amounts of spread are called *heteroscedastic*.
- If the scatter diagram is football-shaped, the normal curve can be used inside a vertical strip, using the conditional mean and SD of y for the given value of x.